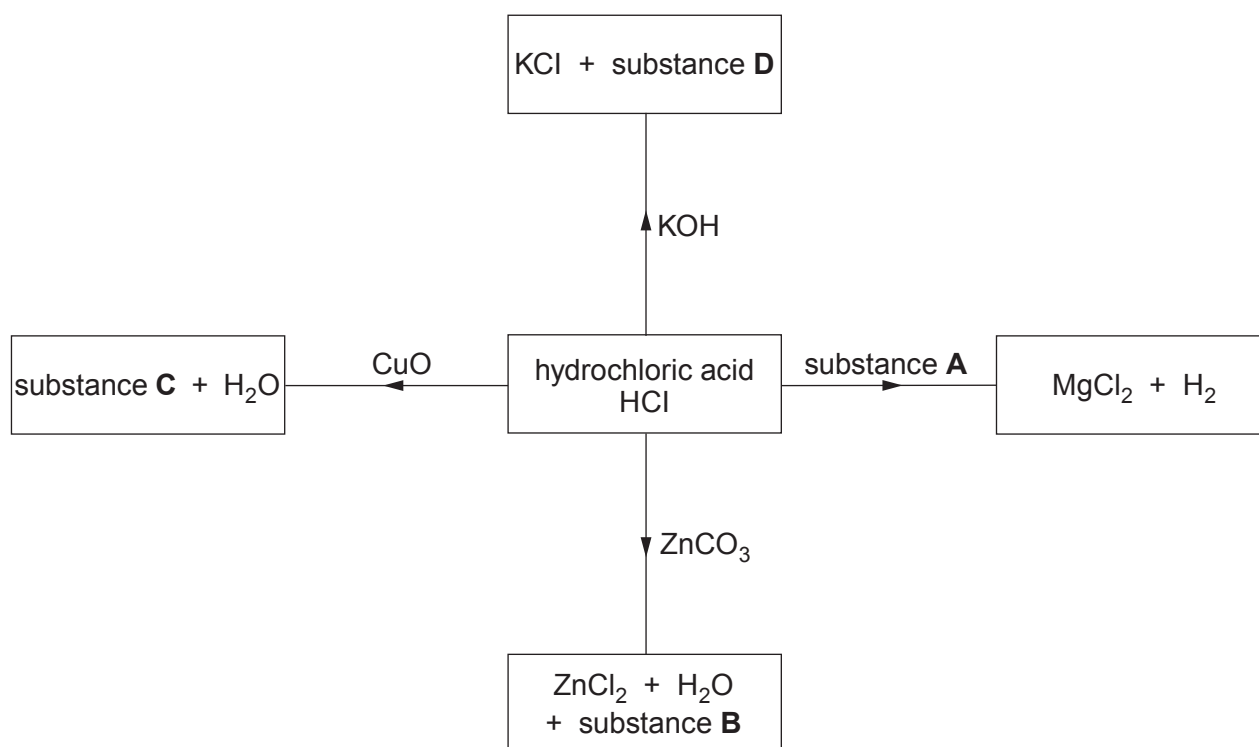


7. The reactions of acids with metals, bases and carbonates are summarised in the following equations.

- acid + metal $\rightarrow$ salt + hydrogen
- acid + base $\rightarrow$ salt + water
- acid + carbonate $\rightarrow$ salt + water + carbon dioxide

(a) The diagram shows some reactions of hydrochloric acid, HCl.



(i) Give the **names** of substances **A** and **B**.

[2]

Substance **A**

Substance **B**

(ii) Give the **formulae** of substances **C** and **D**.

[2]

Substance **C**

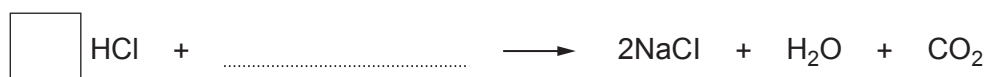
Substance **D**



- (b) Complete the equation for the reaction between hydrochloric acid and sodium carbonate by

- writing the formula of sodium carbonate on the dotted line
- putting a number into the box to balance the equation

[2]

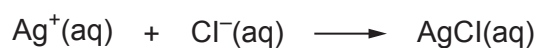
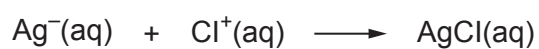
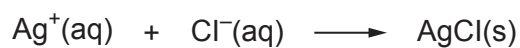
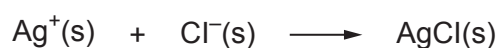
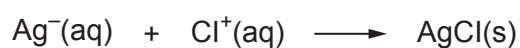


- (c) Silver nitrate solution is used to identify the chloride ions present in hydrochloric acid.

- (i) Give the observation made when silver nitrate solution is added to hydrochloric acid. [1]

.....

- (ii) Put a tick (✓) in the box next to the correct ionic equation for the reaction between silver nitrate and hydrochloric acid. [1]


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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

0

[illegible]

Key

relative atomic mass

atomic number